Applied Energy 228 (2018) 843–851

Contents lists available at ScienceDirect

Applied Energy

journal homepage: www.elsevier.com/locate/apenergy

Enabling strategies of electric vehicles for under frequency load shedding

Hui Liu[a,b], Bin Wang[a,b], Ni Wang[a,b,*], Qiuwei Wu[c], Yude Yang[a,b], Hua Wei[a,b], Canbing Li[d]

[a] College of Electrical Engineering, Guangxi University, Nanning 530004, China
[b] Guangxi Key Laboratory of Power System Optimization and Energy Technology, Guangxi University, Nanning 530004, China
[c] Center for Electric Power and Energy, Department of Electrical Engineering, Technical University of Denmark, Kgs. Lyngby, DK 2800, Denmark
[d] College of Electrical and Information Engineering, Hunan University, Changsha 410082, China

HIGHLIGHTS

- It is the first attempt to develop enabling strategies of EVs for UFLS.
- The hierarchical structure is proposed to enable EVs to take part in UFLS.
- EVs are classified into four grades with different priority levels.
- UFLS-EV, T-UFLS, and A-UFLS are compared.

ARTICLE INFO

Keywords:
Electric vehicle (EV)
Frequency stability
Under-frequency load shedding (UFLS)
Vehicle to grid (V2G)

ABSTRACT

Due to large-scale integration of renewable energy sources into the power grid and the increasing requirements of customers for power quality, under frequency load shedding (UFLS) faces a challenge in ensuring power supply reliability. Large-scale electric vehicles (EVs) as distributed energy storage resources can provide the powerful potential for UFLS to deal with this challenge. In this paper, enabling strategies of EVs are proposed for UFLS to enhance frequency stability and simultaneously reduce power outages, where a hierarchical framework consisting of the control center (CC), EV charging stations (EVCS), and EV terminals (EVT) is developed to implement EVs' participation in UFLS. In the EVCS, the capacity of EVs is calculated based on EV information from the EVT and uploaded to the CC, and at the same time, the dispatch from the CC is distributed among EVs in the EVCS and sent to the EVT. With the proposed strategies, load shedding can be compensated by the vehicle-to-grid power of EVs to reduce power supply interruptions as much as possible. Simulations on the IEEE 9-bus power system have shown the effectiveness of the proposed approach in reducing power outages and enhancing frequency stability.

1. Introduction

In recent years, renewable energy sources (RESs) have been increasing significantly all over the world because of the global environmental concern. By 2020, the penetration of RES generation in the total system generation in China, the United States, and European is expected to reach 27% [1], 26.5% [2], and 20% [3], respectively. The power industry is being restructured whilst a growing number of conventional power plants are being replaced by stochastic RES generation. Under these circumstances, power systems are increasingly operated closer to stability boundaries [4–6]. Ensuring frequency stability, which is the capability of a power system to maintain steady frequency following a severe contingency in generation-load balance [7], is facing a challenge in this situation.

Under frequency load shedding (UFLS) is the most common approach to ensure system frequency stability based on the fast removal of loads during emergencies [8]. In general, UFLS can be divided into the traditional UFLS (T-UFLS) and adaptive UFLS (A-UFLS) [9]. The T-UFLS often causes undesired damages and serious costs due to excessive or inappropriate load removal [10]. This is because the amount of load to be shed in the T-UFLS is determined without taking into account actual fault conditions [11]. Therefore, the A-UFLS is developed considering different fault situations during emergencies. The A-UFLS focuses on the amount of load shedding based on power imbalance and location of load shedding.

The amount of load shedding can be estimated on the basis of the frequency information. In [12], the initial slope of frequency was used to determine the amount of load to be shed considering the reduced-

* Corresponding author at: College of Electrical Engineering, Guangxi University, Nanning 530004, China.
E-mail address: w_n604@163.com (N. Wang).

https://doi.org/10.1016/j.apenergy.2018.06.122
Received 13 April 2018; Received in revised form 14 June 2018; Accepted 24 June 2018
Available online 06 July 2018
0306-2619/ © 2018 Elsevier Ltd. All rights reserved.

Nomenclature

H_i	inertia constant of the ith generator
f_n	rated value of frequency
f_c	frequency of the equivalent inertial center
f_i	frequency at the ith generator bus
ξ	equivalent inertia constant of the entire system
S_1^{Cap}	capacity of EVs in Grade 1
$S_{j,1}^{Cap}$	capacity of EVs in Grade 1 of the jth EVCS
$P_{k,1}^{Cap}$	capacity of the kth EV in Grade 1
$P_{k,1}^{d}$	discharging power of the kth EV in Grade 1
P_{max}^{d}	maximum discharging power
S_2^{Cap}	capacity of EVs in Grade 2
$S_{j,2}^{Cap}$	capacity of EVs in Grade 2 of the jth EVCS
$P_{k,2}^{Cap}$	capacity of the kth EV in Grade 2
$P_{k,2}^{c}$	charging power of the kth EV in Grade 2
S_3^{Cap}	capacity of EVs in Grade 3
$S_{j,3}^{Cap}$	capacity of EVs in Grade 3 of the jth EVCS
$P_{k,3}^{Cap}$	capacity of the kth EV in Grade 3
S_4^{Cap}	capacity of EVs in Grade 4
$S_{j,4}^{Cap}$	capacity of EVs in Grade 4 of the jth EVCS
$P_{k,4}^{Cap}$	capacity of the kth EV in Grade 4
S_i^{Dis}	dispatch for EVs in the ith round of CC
$S_{j,1}^{CS}$	dispatch for EVs in Grade 1 of the jth EVCS
$S_{k,1}^{EV}$	dispatch for the kth EV in Grade 1
$P_{k,1}$	discharging power of an EV in Grade 1
$S_{j,2}^{CS}$	dispatch for EVs in Grade 2 of the jth EVCS
$S_{k,2}^{EV}$	dispatch for the kth EV in Grade 2
$P_{k,2}$	discharging power in Grade 2
$S_{j,3}^{CS}$	dispatch for EVs in Grade 3 of the jth EVCS
$S_{k,3}^{EV}$	dispatch for the kth EV in Grade 3
$P_{k,3}$	discharging power in Grade 3
$S_{j,4}^{CS}$	dispatch for EVs in Grade 4 of the jth EVCS
$S_{k,4}^{EV}$	dispatch for the kth EV in Grade 4
$P_{k,4}$	discharging power in Grade 4
E_k	rated capacity of the kth EV battery
SOC_{k,t_0}	initial battery SOC of the kth EV
$SOC_{k,t}$	real-time battery SOC of the kth EV

order frequency response model. In [13] and [14], the generator swing equation was used to estimate the power imbalance based on frequency deviation and the derivative of frequency. The frequency gradient can also determine the power imbalance under contingency [15]. In order to improve power imbalance estimation, besides system frequency, other factors such as voltage and the rated capacity of generating units are also considered. The load model related to voltage was used to calculate power imbalance on the basis of the swing equation in [16]. The load shedding was determined based on the swing equation, ramp up limit, and rated capacity of generating units [17]. In [18] and [19], the probabilistic UFLS was formulated as a mixed integer linear programming problem to minimize load shedding considering the generator swing equation, governor dynamics, and load damping. The amount of load to be shed was determined by utilizing the spinning reserve of the power system [20]. In [21], an approach based on a few-seconds-in-advance frequency prediction was proposed to determine the amount of load shedding.

Deciding the location of load shedding is another key of A-UFLS. In [13], the sensitivity of the frequency response to the location of disturbances was investigated. The sensitivity of the frequency with respect to load change was used to determine the location and amount of load to be shed in [22]. In [23] and [24], a centralized online algorithm was proposed to determine the location and amount of load shedding based on the magnitude of bus voltages and the static stability margins of bus voltages. In [25], the tracing algorithm of power flow was applied to find the location of load shedding, which can reduce the number of load buses undertaking power imbalance. Voltage stability criterion [26,27] and voltage falling rate [28] were also used to find appropriate location for load shedding. An adaptive UFLS was proposed to determine the optimal location and amount of load shedding by approximating the nonlinearities of swing equations in [29].

A-UFLS can ensure system frequency stability by directly removing the load during contingencies, but it inevitably causes massive power supply interruptions. Therefore, exploring flexible resources has become a 'hot' topic to reduce power supply interruptions as much as possible, while ensuring system frequency stability. With the development of smart energy management at the demand side [30], flexible resources such as smart appliances [31], electric vehicles (EVs), and energy storage systems [32] are used to improve frequency stability. In [23], smart appliances were enabled to implement UFLS during emergencies. The water heaters were used to provide power support to the power grid during the UFLS [33]. In [34], a method was proposed for the energy storage system to prevent load shedding during transient declines of frequency. EVs have shown the powerful potential for regulation due to their vehicle-to-grid (V2G) capability as distributed energy storage units [35–37]. In [38], the coordinated control between EVs and conventional generating units was proposed for EVs to participate in load frequency control. In [39] and [40], decentralized V2G control strategies were proposed to perform primary frequency control taking charging behaviors into consideration. Droop charging control was proposed for an EV aggregator to enhance frequency stability in a microgrid with the high penetration of RESs [41]. Secondary frequency regulation strategies were addressed considering charging demands of EVs in [42], the tradeoff between charging facilities and energy storage devices in [43], and driving demands and aggregator's benefits in [44]. The primary frequency control with EVs' participation can improve the quality of system frequency and can also postpone UFLS [45]. However, the existing work mentioned above focuses on frequency control from the point of view of power system steady states. Although EVs can also provide the support for ensuring frequency stability during emergencies as in [46], control strategies responding to UFLS instruction during contingencies are missing.

In this paper, adaptive UFLS strategies with EVs' participation (UFLS-EV) are developed to ensure the frequency stability of power systems during emergencies. The contributions can be summarized as follow.

- It is the first attempt to develop enabling strategies of EVs for performing UFLS.
- The hierarchical structure is proposed to enable EVs to take part in UFLS, which consists of the control center (CC), EV charging stations (EVCS), and EV terminals (EVT).
- EVs are classified into four grades with different priority levels to reduce the influence of UFLS on EV battery energy as much as possible.
- UFLS-EV, T-UFLS, and A-UFLS are compared in terms of restoring system frequency and reducing power outages.

The remainder of this paper is as follows. In Section 2, the hierarchical framework of UFLS-EV is addressed. Power imbalance estimation and the capacity of EVs are discussed in Section 3. The hierarchical control strategies of UFLS-EV are proposed in Section 4. The simulation system is provided in Section 5. Simulations and discussions are performed in Section 6, followed by conclusions.

2. The hierarchical framework of UFLS-EV

The UFLS has to face power outages due to load shedding, which causes massive economic losses. EVs as distributed energy storage units can reduce power outages by discharging the power to the grid or decreasing charging as much as possible. Our motivations of proposing UFLS-EV stem from the following four advantages.

(1) Fast charging/discharging characteristics of EVs. EVs are integrated into the power grid through power electronic devices with the response time of a millisecond level, which can enable the fast switching between the charging and discharging of EVs to restrain frequency drop during emergencies. Note that this switching only occurs while the power system is subjected to an extremely severe fault, which means that EV battery life will not be shorten.
(2) The significant capacity of EVs available for emergency control. During emergencies, UFLS as the last defense to ensure frequency stability usually compensates the generation-load mismatch by load shedding. However, the large-scale integration of EVs into the power grid can provide a significant support for this mismatch instead of load shedding.
(3) Enhancing power supply reliability. During emergencies, power outages inevitably occur due to the load shedding in T-UFLS and A-UFLS. The UFLS-EV can reduce the power outages by instantly releasing the power to the grid during emergencies instead of load shedding.
(4) Small negative impacts on EV battery energy. The process of UFLS generally lasts tens of seconds, which means that EV battery energy levels will not be changed significantly, even if EVs discharge with maximum power or stop charging.

The hierarchical framework of UFLS-EV in Fig. 1 consists of EVT, EVCS, and CC.

In the EVT, EV information, including charging/discharging power, battery SOC levels, plug-in time, plug-out time, and the expected charging, is uploaded to the EVCS, and at the same time, the instruction from the EVCS is implemented by regulating the charging/discharging of EVs.

In the EVCS, the capacity of EVs is calculated and uploaded to the CC. Besides, the dispatch from the CC is distributed among EVs and sent to the EVT.

In the CC, the capacity uploaded by the EVCS is summed up. At the same time, the dispatch to the EVCS is made within the capacity of EVs based on frequency information and generator states acquired through wide area measurement systems (WAMS).

3. Power imbalance estimation and the capacity of EVs

While the frequency fast falls below the certain level which activates the UFLS, the power imbalance of a system will be first estimated for performing load shedding in UFLS. Before load shedding is dispatched to EVs, the capacity of EVs has to be calculated for the UFLS.

3.1. Power imbalance estimation

In a multi-machine power system, the power imbalance can be calculated according to (1) based on the frequency information at generator buses, which is extensively used to estimate the power imbalance in the literature [11].

$$\Delta P = \sum \Delta P_i = \frac{2\sum H_i}{f_n}\frac{df_c}{dt} = \xi\frac{df_c}{dt} \tag{1}$$

where

$$f_c = \sum H_i f_i / \sum H_i,\ \xi = \frac{2}{f_n}\sum H_i$$

3.2. The capacity of EVs for UFLS

Charging demands vary from EV to EV due to different transport usages. For example, an EV with a high SOC level can maintain battery energy, while an EV with a low SOC level has to be charged for satisfying next trip. To reduce the influence of the UFLS on scheduled charging of EVs, although the influence is small as shown in Section 2.1, EVs can be classified into three types as follows.

- Selling electricity (called TYPE I): Because the battery energy of EVs is high enough for transport usages, users want to sell the redundant energy to the power grid for the revenue. EVs in TYPE I can be considered as the power supply available for the UFLS during emergencies.
- Charging EVs (called TYPE II): EVs have to be charged to satisfy users' charging demands. Decreasing the charging power even stopping charging can reduce the power imbalance during emergencies. EVs in this type can also release the power to the grid during the UFLS.
- Maintaining battery energy (called TYPE III): EVs may remain their battery energy unchanged, because the battery energy is high enough and users would like to charge their EVs at night. EVs in TYPE III can provide the reserve for the grid during emergencies.

The three types of EVs can be divided into the following four grades with different priority levels.

(1) EVs in Grade 1 with the highest priority

EVs in TYPE I are ranked as Grade 1 with the highest priority while taking part in UFLS, because the battery energy is high enough and EV users would like to sell the electricity to the power grid. The capacity of EVs illustrated in Fig. 2 can be calculated by (2).

$$\begin{cases} S_1^{\mathrm{Cap}} = \sum S_{j,1}^{\mathrm{Cap}} \\ S_{j,1}^{\mathrm{Cap}} = \sum P_{k,1}^{\mathrm{Cap}} \\ P_{k,1}^{\mathrm{Cap}} = P_{\max}^{\mathrm{d}} - P_{k,1}^{\mathrm{d}} \end{cases} \tag{2}$$

(2) EVs in Grade 2 with the second priority

EVs in TYPE II can stop charging to reduce the generation-load mismatch, and thus they are ranked as Grade 2 with the secondary priority. The capacity of EVs is illustrated in Fig. 2 and calculated as

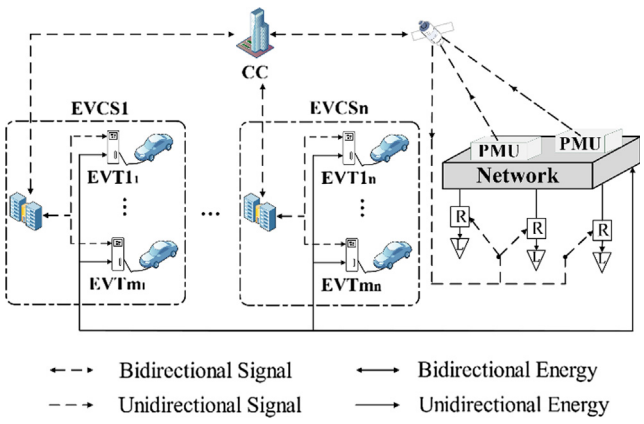


Fig. 1. The hierarchical framework of UFLS-EV.

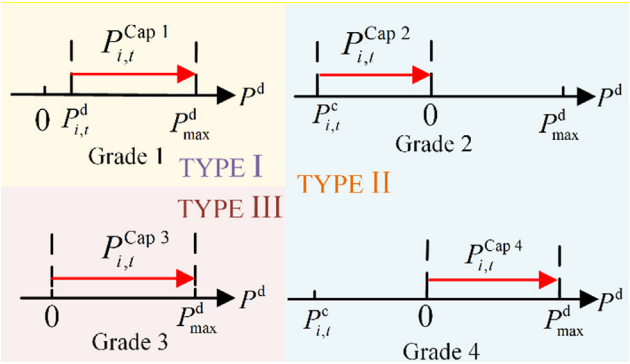


Fig. 2. The capacity of EVs in all Grades for UFLS.

$$\begin{cases} S_2^{\mathrm{Cap}} = \sum S_{j,2}^{\mathrm{Cap}} \\ S_{j,2}^{\mathrm{Cap}} = \sum P_{k,2}^{\mathrm{Cap}} \\ P_{k,2}^{\mathrm{Cap}} = -P_{k,2}^{\mathrm{c}} \end{cases} \tag{3}$$

(3) EVs in Grade 3 with the third priority

EVs in TYPE III are ranked as Grade 3 with the third priority. The capacity of EVs illustrated in Fig. 2 can be calculated by (4).

$$\begin{cases} S_3^{\mathrm{Cap}} = \sum S_{j,3}^{\mathrm{Cap}} \\ S_{j,3}^{\mathrm{Cap}} = \sum P_{k,3}^{\mathrm{Cap}} \\ P_{k,3}^{\mathrm{Cap}} = P_{\max}^{\mathrm{d}} \end{cases} \tag{4}$$

(4) EVs in Grade 4 with the last priority

While the first three grades of EVs cannot compensate the generation-load mismatch during emergencies, EVs in TYPE II (i.e. EVs in Grade 2) have to release the power to the grid and are ranked as Grade 4 with the last priority. The capacity of EVs shown in Fig. 2 can be calculated by (5)

$$\begin{cases} S_4^{\mathrm{Cap}} = \sum S_{j,4}^{\mathrm{Cap}} \\ S_{j,4}^{\mathrm{Cap}} = \sum P_{k,4}^{\mathrm{Cap}} \\ P_{k,4}^{\mathrm{Cap}} = P_{\max}^{\mathrm{d}} \end{cases} \tag{5}$$

4. Adaptive hierarchical control strategies for UFLS-EV

4.1. Adaptive hierarchical control strategies

Once the frequency fast falls below the predetermined threshold during emergencies, the power imbalance is estimated according to (1), and then the load shedding is performed round by round. The load shedding in each round is dispatched from the CC to the EVCS and distributed among EVs in the EVCS according to the following strategies.

(1) Dispatch of EVs in Grade 1

When the capacity of EVs in Grade 1 can cover the load shedding in the current round, only EVs in Grade 1 will perform the UFLS with the following hierarchical control.

In the CC, the load shedding of EVs is dispatched to the EVCS according to (6).

$$S_{j,1}^{\mathrm{CS}} = \frac{S_{j,1}^{\mathrm{Cap}}}{S_1^{\mathrm{Cap}}} \cdot S_i^{\mathrm{Dis}} \tag{6}$$

In the EVCS, load shedding is allocated among EVs in Grade 1 with the following expression.

$$S_{k,1}^{\mathrm{EV}} = \frac{P_{k,1}^{\mathrm{Cap}}}{S_{j,1}^{\mathrm{Cap}}} \cdot S_{j,1}^{\mathrm{CS}} \tag{7}$$

In the EVT, the V2G power of an EV can be shown as,

$$P_{k,1} = P_{k,1}^{\mathrm{d}} + S_{k,1}^{\mathrm{EV}} \tag{8}$$

(2) Dispatch of EVs in Grade 2

When the capacity of EVs in Grade 1 is not enough for performing the load shedding in the current round, EVs in Grade 2 have to participate in the UFLS.

In the CC, the load shedding of EVs in Grade 2 is dispatched to the EVCS as

$$S_{j,2}^{\mathrm{CS}} = \frac{S_{j,2}^{\mathrm{Cap}}}{S_2^{\mathrm{Cap}}} \cdot (S_i^{\mathrm{Dis}} - S_1^{\mathrm{Cap}}) \tag{9}$$

In the EVCS, the load shedding of EVs in Grade 2 is allocated according to (10).

$$S_{k,2}^{\mathrm{EV}} = \frac{P_{k,2}^{\mathrm{Cap}}}{S_{j,2}^{\mathrm{Cap}}} \cdot S_{j,2}^{\mathrm{CS}} \tag{10}$$

In the EVT, in this round, the V2G power of EVs in Grades 1 and 2 can be expressed as,

$$\begin{cases} P_{k,1} = P_{\max}^{\mathrm{d}} \\ P_{k,2} = P_{k,2}^{\mathrm{c}} + S_{k,2}^{\mathrm{EV}} \end{cases} \tag{11}$$

(3) Dispatch of EVs in Grade 3

If the capacity of EVs in the first two Grades is not enough for the load shedding of the current round, EVs in Grade 3 are activated to perform the UFLS.

In the CC, the load shedding of EVs in Grade 3 is sent to the EVCS according to (12):

$$S_{j,3}^{\mathrm{CS}} = \frac{S_{j,3}^{\mathrm{Cap}}}{S_3^{\mathrm{Cap}}} \cdot (S_i^{\mathrm{Dis}} - S_1^{\mathrm{Cap}} - S_2^{\mathrm{Cap}}) \tag{12}$$

In the EVCS, the load shedding of EVs in Grade 3 is calculated in the following form,

$$S_{k,3}^{\mathrm{EV}} = \frac{P_{k,3}^{\mathrm{Cap}}}{S_{j,3}^{\mathrm{Cap}}} \cdot S_{j,3}^{\mathrm{CS}} \tag{13}$$

In the EVT, the V2G power of EVs in Grades 1, 2, and 3 can be expressed as,

$$\begin{cases} P_{k,1} = P_{\max}^{\mathrm{d}} \\ P_{k,2} = 0 \\ P_{k,3} = S_{k,3}^{\mathrm{EV}} \end{cases} \tag{14}$$

(4) Dispatch of EVs in Grade 4

When EVs in Grade 4 participate in the UFLS of this round, the dispatch from CC to EVCS is shown as follows.

In the CC, the load shedding of EVs in Grade 4 is sent to the EVCS following the following expression.

$$S_{j,4}^{\mathrm{CS}} = \frac{S_{j,4}^{\mathrm{Cap}}}{S_4^{\mathrm{Cap}}} \cdot (S_i^{\mathrm{Dis}} - S_1^{\mathrm{Cap}} - S_2^{\mathrm{Cap}} - S_3^{\mathrm{Cap}}) \tag{15}$$

In the EVCS, the load shedding of EVs in Grade 4 is expressed in

(16).

$$S_{k,4}^{EV} = \frac{P_{k,4}^{Cap}}{S_{j,4}^{Cap}} \cdot S_{j,4}^{CS} \tag{16}$$

In the EVT, the V2G power of EVs in three types can be expressed as,

$$\begin{cases} P_{k,1} = P_{max}^{d} \\ P_{k,2} = S_{k,4}^{EV} \\ P_{k,3} = P_{max}^{d} \end{cases} \tag{17}$$

4.2. The flowchart of UFLS-EV

The flowchart of UFLS-EV is shown in Fig. 3 with the following steps.

Step 1: The system frequency is monitored through WAMS based on PMU in the CC, as shown in Fig. 1.

Step 2: If the frequency is below a given threshold value, continue **Step 3**; or return to **Step 1**.

Step 3: The power imbalance is estimated according to (1), and load shedding parameters including frequency thresholds and time delay in each round are set in the CC.

Step 4: In the *i*th round load shedding, the capacity of EVs in each grade is calculated according to (2)-(5) in the EVCS and uploaded to the CC.

Step 5: According to the capacity of EVs, the load shedding in the *i*th round is dispatched to EVs as much as possible. If the capacity of EVs is not enough, load removal has to occur.

Step 6: The task for EVs is dispatched to the EVCS according to (6), (9), (12), and (15) in the CC and distributed among EVs according to (7), (10), (13), and (16) in the EVCS.

Step 7: In the EVCS, the control command is sent to the EVT to control the charging/discharging of EVs according to (8), (11), (14), and (17).

Step 8: If the system frequency returns to the allowable range, the UFLS stops; or continue to perform **Step 4**.

5. Simulation system

5.1. Power system model

In this simulation, the IEEE 9-bus system (please see [47] for the detailed information) in Fig. 4 is used, where the generators connected to buses 1, 2, and 3 are modeled by the two-order classical model, and the load at buses 5, 6, and 8 is represented with constant impedances.

5.2. Electric vehicle model

During the charging/discharging process, the variation of an EV battery energy can be expressed in (18) [39], and the parameters of EVs are listed in Table 1.

$$SOC_{k,t} = SOC_{k,t_0} - \frac{1}{E_k}\int_{t_0}^{t} P_{k,t}\,dt \tag{18}$$

In order to prevent EV batteries from the over-charging or -discharging, maximum and minimum SOC levels are considered.

6. Simulation and discussion

Simulations under MATLAB/SIMULINK environment are used to examine the effectiveness of the proposed UFLS-EV strategies.

6.1. Simulation scenario

Seven charging stations which manage 5000 EVs are assumed to connect to buses 5, 6, and 8, respectively, each of which has three types of EVs as shown in Table 2. As shown in Table 3, the simulation parameters of EVs, including initial battery SOC, initial charging/discharging power, and the expected battery SOC, follow a normal distribution imitated by the Monte Carlo simulation method. The capacity of EVs in four Grades of seven EVCSs is shown in Fig. 5. Note that an EV will not participate in UFLS while its battery energy is out of the defined range shown in Table 1.

To show the effectiveness of the adaptive UFLS-EV strategies proposed in this paper, the T-UFLS, A-UFLS, and UFLS-EV are simulated and compared.

As shown in Table 4, the UFLS includes five basic rounds and three special rounds, in each round of which, frequency thresholds, delay time, and the amount of load to be shed are set according to [9]. For the T-UFLS, 25% of current load is considered as the load to be shed, while for the A-UFLS and UFLS-EV, the power imbalance is calculated in real time according to (1).

6.2. Simulation and comparisons

Two scenarios are used to show the advantages of the UFLS-EV over T-UFLS and A-UFLS in reducing load shedding and restoring system frequency.

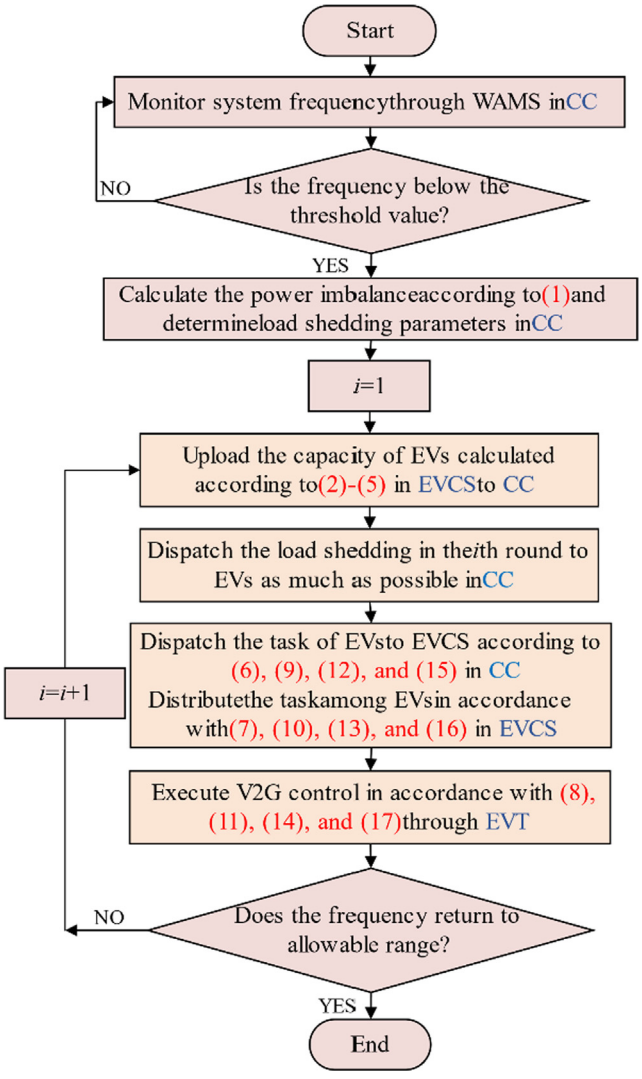


Fig. 3. The flowchart of UFLS-EV.

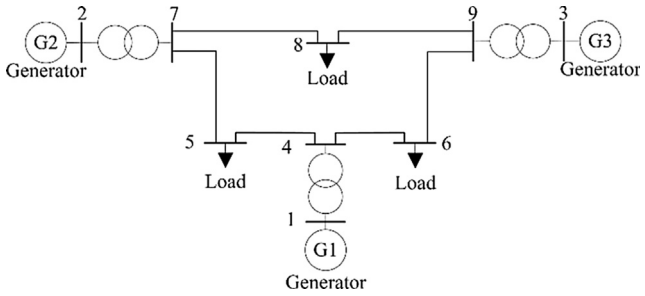


Fig. 4. The IEEE 9-bus power system.

Table 1
The V2G parameters of EVs.

Parameters	Value
Battery capacity (kWh)	32
Charging/discharging efficiency	0.92/0.92
Maximum power (kW)	7
Maximum/minimum SOC(p.u.)	0.9/0.1

Table 2
The parameters of EV charging stations.

Bus	EVCS	TYPE 1	TYPE 2	TYPE 3
5	I	136	477	68
	II	105	367	53
	III	156	545	77
6	IV	149	522	74
	V	137	479	68
8	VI	161	564	81
	VII	156	547	78

Table 3
Simulation parameter of EVs.

	TYPE 1	TYPE 2	TYPE 3
Initial SOC (p.u.)	SOC ~ N (0.7,0.01) SOC ∈ [0.6, 0.8]	SOC ~ N (0.4,0.01) SOC ∈ [0.3, 0.5]	SOC ~ N (0.7,0.01) SOC ∈ [0.6, 0.8]
Expected SOC (p.u.)	SOC ~ N (0.4,0.01) SOC ∈ [0.3, 0.5]	SOC ~ N (0.7,0.01) SOC ∈ [0.6, 0.8]	
Initial power (kW)	P ~ N (3.5,1) P∈[7,0]	P ~ N (−3.5,1) P ∈ [−7, 0]	0

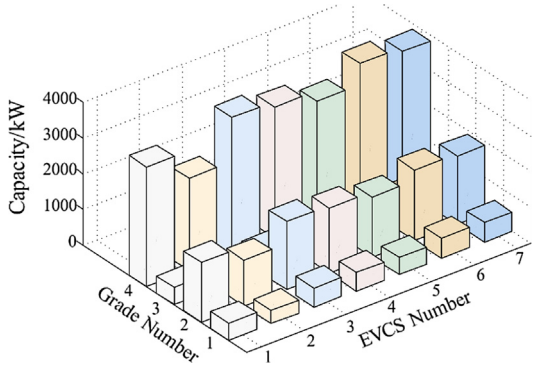


Fig. 5. The capacity in four Grades of seven EVCSs.

(1) **Case 1: Generator trip**

When the generator connected to bus 3 is disconnected at $t = 10$ s, the power imbalance is 85 MW about 26.59% of system current load.

Table 4
Parameter setting of UFLS.

Round		Frequency setting (Hz)	Time delay (s)	Load to be shed (ΔP%)
Basic round	1	49.30	0.2	25
	2	49.10	0.2	25
	3	48.90	0.2	20
	4	48.70	0.2	15
	5	48.50	0.2	15
Special round	1	49.50	15	8
	2	49.50	20	8
	3	49.50	25	8

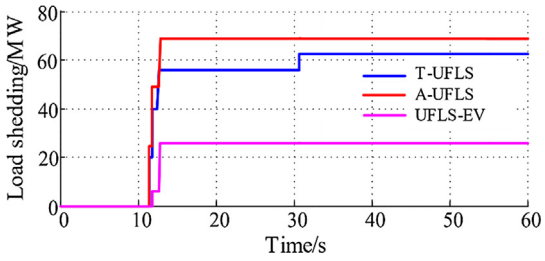


Fig. 6. Load removal for T-UFLS, A-UFLS, and UFLS-EV in Case 1.

The load removal for T-UFLS, A-UFLS, and UFLS-EV is shown in Fig. 6.

With the T-UFLS, the first three basic rounds act at $t = 11.43$ s, $t = 11.79$ s, and $t = 12.38$ s, for each of which, load shedding is 19.97 MW, 19.98 MW, and 15.98 MW, respectively. A special round acts at $t = 30.75$ s with the load shedding of 6.39 MW. The load of 62.32 MW is removed by the T-UFLS. For the A-UFLS, three basic rounds are activated, as shown in Fig. 6, for each of which, the load shedding is 24.78 MW, 24.78 MW, and 19.83 MW, respectively. This means that the load of 69.39 MW has to be removed by the A-UFLS. However, for the UFLS-EV, the load of 25.71 MW is only removed while the capacity of EVs is exhausted, as shown in Fig. 6. Therefore, power outages are greatly reduced by the UFLS-EV compared with the T-UFLS and A-UFLS.

For the UFLS-EV shown in Fig. 7, in the first round, the capacity of EVs in Grade 1, Grade 2, and Grade 3 are depleted with 3.46 MW, 12.21 MW, and 3.50 MW, respectively, and the load shedding of 5.60 MW is undertaken by EVs in Grade 4. In the second round, the load of 5.89 MW has to be removed, because the capacity of 18.90 MW in Grade 4 cannot compensate the load shedding of 24.78 MW in this round. Therefore, the power of 43.68 MW is provided by EVs to replace load removal, which shows that the load of 43.68 MW will maintain power supply.

For clarity, we randomly choose an EV in each of three types for demonstration, as shown in Fig. 8. In the first round, EVs in TYPE I and TYPE III release the power with the maximum value, and an EV in TYPE II stops charging and then discharges with the power of 1.60 kW. In the second round, only an EV in TYPE II can provide the support for the grid with the maximum power. Therefore, in the second round, the

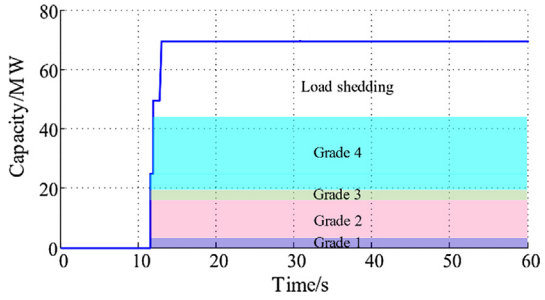


Fig. 7. The V2G power support provided by EVs in Case 1.

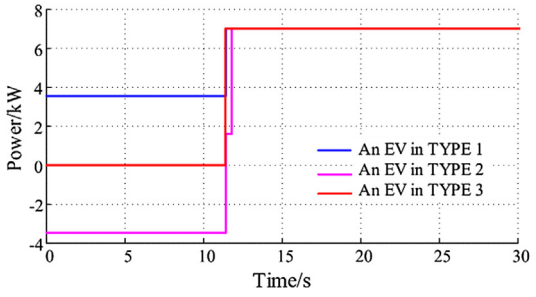


Fig. 8. The V2G power of three EVs in Case 1.

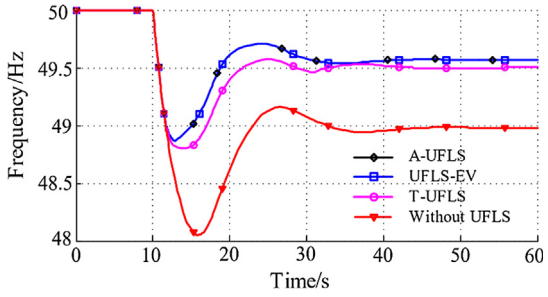


Fig. 9. The response of system frequency in Case 1.

Table 5
Minimum frequency and steady frequency in Case 1.

	Minimum frequency (Hz)	Steady frequency (Hz)
Without UFLS	48.05	48.98
T-UFLS	48.80	49.50
A-UFLS	48.87	49.57
UFLS-EV	48.87	49.57

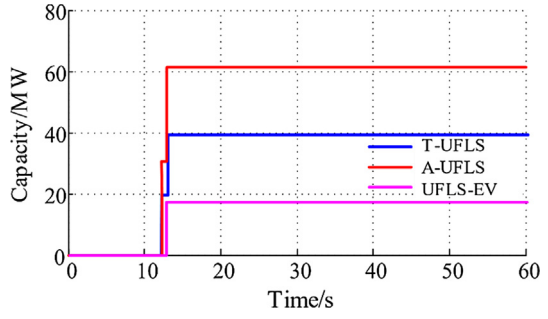


Fig. 10. Load removal for T-UFLS and UFLS-EV in Case 2.

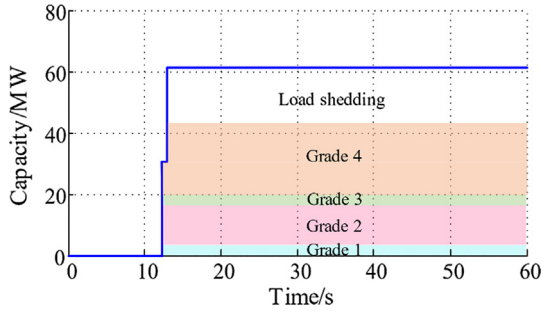


Fig. 11. The V2G power support provided by EVs in Case 2.

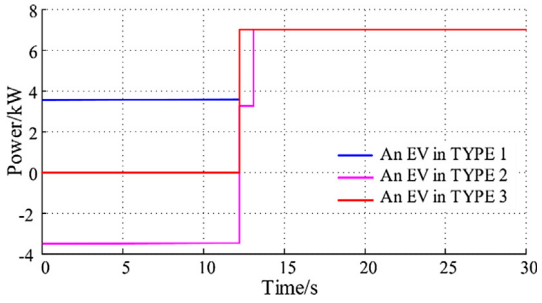


Fig. 12. The V2G power of three EVs in Case 2.

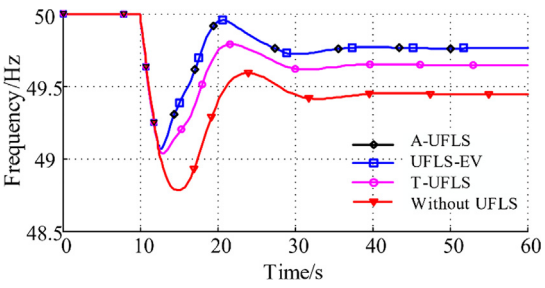


Fig. 13. The response of system frequency in Case 2.

Table 6
Minimum frequency and steady frequency in Case 2.

	Minimum frequency (Hz)	Steady frequency (Hz)
Without UFLS	48.62	49.37
T-UFLS	48.91	49.56
A-UFLS	48.94	49.60
UFLS with EV	48.94	49.60

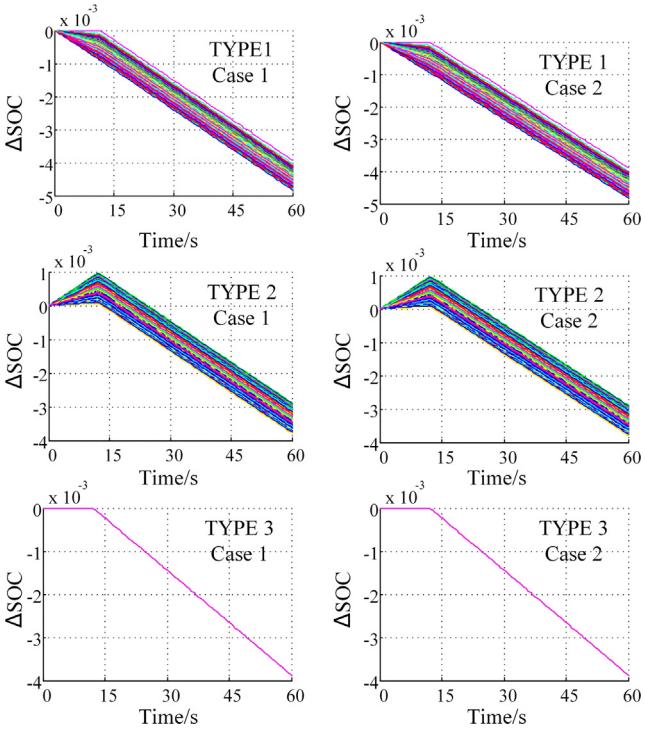


Fig. 14. Impacts of UFLS-EV on battery SOC levels.

capacity of EVs is exhausted.

For the T-UFLS, A-UFLS, and UFLS-EV, the fast fall of system frequency can be suppressed, as shown in Fig. 9. Compared with T-UFLS, better frequency recovery can be achieved by the A-UFLS and UFLS-EV in terms of minimum frequency and steady frequency, as illustrated in Table 5. This is because the power imbalance is estimated more accurately by the A-UFLS and UFLS-EV than the T-UFLS. For the A-UFLS and UFLS-EV, the frequency recovery has the same dynamics illustrated in Fig. 9, because the same power imbalance is used.

Therefore, in the case that the power system is subjected to a generator trip, the three strategies can restore the frequency to the predetermined range, and the A-UFLS and UFLS-EV can better achieve frequency recovery than the T-UFLS. However, a large number of load has to be removed by the T-UFLS and A-UFLS, while load shedding is reduced by the UFLS-EV due to the compensation of V2G power of EVs.

(2) **Case 2: Load increase**

The load of 102.29 MW is suddenly increased at $t = 10$ s, which is about 32% of system current load. The load removal is illustrated in Fig. 10 for the T-UFLS, A-UFLS, and UFLS-EV.

By the T-UFLS, two basic rounds act at $t = 11.96$ s and $t = 12.61$ s with the load removal of 20 MW and 19.95 MW, respectively. The power outages of 39.95 MW inevitably occur. For the A-UFLS, two basic rounds act with the load removal of 30.60 MW and 30.60 MW, respectively. Therefore, A-UFLS causes the power outages of 61.20 MW, as shown in Fig. 10. However, for the UFLS-EV, the load removal is reduced to a great extent due to the compensation of EVs.

For the UFLS-EV, two basic rounds act, and the V2G power support during UFLS can be shown in Fig. 11. In the first round, the capacity of EVs in Grade 1, Grade 2, and Grade 3 is exhausted with 3.46 MW, 12.21 MW, and 3.50 MW, and EVs in Grade 2 (i.e. Grade 4) will continue to release the power with the capacity of 11.42 MW. In the second round, the capacity of EVs in Grade 4 is depleted. With the two basic rounds, EVs can provide the power of 43.68 MW to compensate the generation-load mismatch during emergencies. Therefore, the power outages of 43.67 MW are avoided.

We still take an EV from each of three types as an example. The V2G power of the three EVs is shown in Fig. 12. In the first round, the capacity of all EVs is exhausted except for an EV in TYPE II (i.e., Grade 4), as shown in Fig. 12. In the second round, an EV in Grade 4 will release the power to the grid with the maximum value.

For the T-UFLS, A-UFLS, and UFLS-EV, the frequency recovery is illustrated in Fig. 13. Compared to the T-UFLS, better minimum frequency and steady frequency can be achieved by the A-UFLS and UFLS-EV as shown in Table 6, because the power imbalance can be estimated more accurately for the A-UFLS and UFLS-EV.

Therefore, for Case 2, the UFLS-EV shows the more advantages over the T-UFLS and A-UFLS in terms of improving frequency stability and reducing load removal.

We can also show the impact of the UFLS-EV on battery SOC levels in Fig. 14. Although the battery SOC levels of EVs deviate from the scheduled due to implementing the UFLS, this influence is small. For Case 1 and Case 2, EV battery SOC levels are reduced below 0.5%. This is because the UFLS generally lasts tens of seconds, as addressed in Section 2.1. Therefore, the UFLS-EV does not affect the scheduled charging of EVs to a great extent.

7. Conclusion

In this paper, it is the first attempt to propose adaptive UFLS-EV strategies to enable the participation of EVs in UFLS. For the UFLS-EV, a hierarchical framework is developed to perform the dispatch from CC to EVs based on the information exchange among CC, EVCS, and EVT. With the UFLS-EV, system frequency stability can be ensured, and load removal is reduced to a great extent. Besides, there is a small negative impact on EV battery SOC levels, because UFLS generally lasts tens of seconds. Simulations on the IEEE 9-bus power system demonstrate the effectiveness of the proposed UFLS-EV strategies in performing the UFLS. Compared with the T-UFLS, besides that load removal can be reduced significantly, system frequency can be restored to a higher level by the UFLS-EV. Although both the A-UFLS and UFLS-EV achieve the same frequency dynamics, the UFLS-EV can reduce power outages as much as possible while ensuring frequency stability.

The UFLS-EV can be applied to prevent the frequency of the power system from collapse by the compensation of EVs.

Acknowledgment

This work was supported by the National Natural Science Foundation of China (Grant No. 51577085) and by the Scientific Research Foundation of Guangxi University (Grant No. XTZ160522).

References

[1] China National Renewable Energy Centre. “China Renewable Energy outlook 2017, ” Beijing; 2017.
[2] Federal Energy Regulatory Commission. “Energy Infrastructure Update For November 2017,” Washington, D.C.; 2017.
[3] Communication from the Commission to the European Parliament and the Council. “Renewable energy: Progressing towards the 2020 target,” Brussels, COM; 2011.
[4] Osamaa A, El-Said SA, Hassanien AE. Handbook of research on emerging technologies for electrical power planning, analysis, and optimization, Pennsylvania; 2016.
[5] Chang-Chien LR, An LN, Lin T, Lee W. Incorporating demand response with spinning reserve to realize an adaptive frequency restoration plan for system contingencies. IEEE Trans Smart Grid 2012;3(3):1145–53.
[6] Saffarian A, Sanaye-Pasand M. Enhancement of power system stability using adaptive combinational load shedding methods. IEEE Trans Power Syst 2011;26(3):1010–20.
[7] Kundur P, Paserba J, Ajjarapu V, et al. Definition and classification of power system stability IEEE/CIGRE joint task force on stability terms and definitions. IEEE Trans Power Syst 2004;19(3):1387–401.
[8] Kundur P. Power system stability and control. New York: McGraw-Hill; 1994.
[9] IEEE Power Engineering Society: IEEE Guide for the Application of Protective Relays Used for Abnormal Frequency Load Shedding and Restoration. IEEE Std. C37.117TM-2007, Aug. 24, 2007.
[10] Laghari JA, Mokhlis H, Bakar AHA, Mohamad H. Application of computational intelligence techniques for load shedding in power systems: a review. Energy Convers Manage 2013;75:130–40.
[11] Ketabi A, Fini MH. An underfrequency load shedding scheme for hybrid and multiarea power systems. IEEE Trans Smart Grid Jan. 2015;6(1):82–91.
[12] Anderson PM, Mirheyder M. An, adaptive method for setting under frequency load shedding relays. IEEE Trans Power Syst 1992;7(2):647–55.
[13] Terzija VV. Adaptive under frequency load shedding based on the magnitude of the disturbance estimation. IEEE Trans Power Syst 2006;21(3):1260–6.
[14] Abdelwahid S, Babiker A, Eltom A, et al. Hardware implementation of an automatic adaptive aentralized underfrequency load shedding scheme. IEEE Trans Power Delivery 2013;29(6):2664–73.
[15] Rudez U, Mihalic R. Analysis of under frequency load shedding using a frequency gradient. IEEE Trans Power Delivery Apr. 2011;26(2):565–75.
[16] Rudez U, Mihalic R. Monitoring the first frequency derivative to improve adaptive under frequency load-shedding schemes. IEEE Trans Power Syst 2011;26(2):839–46.
[17] Tofis Y, Timotheou S, Kyriakides E. Minimal load shedding using the swing equation. IEEE Trans Power Syst May. 2016;32(3):2466–7.
[18] Amraee T, Darebaghi MG, Soroudi A, et al. Probabilistic under frequency load shedding considering RoCoF relays of distributed generators. IEEE Trans Power Syst [early access].
[19] Darebaghi MG, Amraee T. Dynamic multi-stage under frequency load shedding considering uncertainty of generation loss. IET Gener Transm Distrib Jul. 2017;11(13):3202–9.
[20] Jallad J, Mekhilef S, Mokhlis H, et al. Improved UFLS with consideration of power deficit during shedding process and flexible load selection. IET Renew Power Gener 2018;12(5):565–75.
[21] Rudez U, Mihalic R. WAMS-based underfrequency load shedding with short-term frequency prediction. IEEE Trans Power Delivery 2016;31(4):1912–20.
[22] Reddy CP, Chakrabarti S, Srivastava SC. A sensitivity-based method for under-frequency load-shedding. IEEE Trans Power Syst Mar. 2014;29(2):984–5.
[23] Wang J, Zhang H, Zhou Y. Intelligent under frequency and under voltage load shedding method based on the active participation of smart appliances. IEEE Trans Smart Grid 2017;8(1):353–61.
[24] Seyedi H, Sanaye-Pasand M. New centralized adaptive load shedding algorithms to mitigate power system blackouts. IET Gener Transm Distrib 2009;3(1):99–114.
[25] Tang J, Liu J, Ponci F, Monti A. Adaptive load shedding based on combined frequency and voltage stability assessment using synchrophasor measurements. IEEE Trans Power Syst 2013;28(2):2035–47.

[26] Shekari T, Aminifar F, Sanaye-Pasand M. An analytical adaptive load shedding scheme against severe combinational disturbances. IEEE Trans Power Syst 2016;31(5):4135–43.
[27] Seethalekshmi K, Singh SN, Srivastava SC. A synchrophasor assisted frequency and voltage stability based load shedding scheme for self-healing of power system. IEEE Trans Smart Grid 2011;2(2):221–30.
[28] Abedini M, Sanaye-Pasand M, Azizi S. Adaptive load shedding scheme to preserve the power system stability following large disturbances. IET Gener Transm Distrib 2014;8(12):2124–33.
[29] Tofis Y, Yiasemi Y, Kyriakides E. A plug-and-play selective load shedding scheme for power systems. IEEE Syst J 2017;11(4):1–8.
[30] Eissa MM. First time real time incentive demand response program in smart grid with 'i-Energy' management system with different resources. Appl Energy Feb. 2018;212:607–21.
[31] Lakshmanan V, Marinelli M, Hu J, Bindner HW. Provision of secondary frequency control via demand response activation on thermostatically controlled loads: solutions and experiences from Denmark. Appl Energy Jul. 2016;173:470–80.
[32] Li J, Xiong R, Yang Q, Liang F, Zhang M, Yuan W. Design/test of a hybrid energy storage system for primary frequency control using a dynamic droop method in an isolated microgrid power system. Appl Energy 2017;201:207–69.
[33] Qi Y, Wang D, Jia H, Chen N, Wei W. Research on under frequency load shedding strategy using aggregated thermostatically controlled loads based on demand response. Proc CSEE Feb 2017;37(3):751–60.
[34] Pulendran S, Tate JE. Energy storage system control for prevention of transient under-frequency load shedding. IEEE Trans Smart Grid 2017;8(2):927–36.
[35] Liu H, Qi J, Wang J, Li P, Li C, Wei H. EV Dispatch control for supplementary frequency regulation considering the expectation of EV owners. IEEE Trans Smart Grid [early access].
[36] Liu H, Huang K, Yang Y, Wei H, Ma S. Real-time vehicle-to-grid control for frequency regulation with high frequency regulating signal. Protect Cont Modern Power Syst 2018;3(3):141–8.
[37] Cai H, Chen Q, Guan Z, Huang J. Day-ahead optimal charging/ discharging scheduling for electric vehicles in microgrids. Protect Cont Modern Power Syst 2018;3(3):93–107.
[38] Jia H, Li X, Mu Y, et al. Coordinated control for EV aggregators and power plants in frequency regulation considering time-varying delays. Appl Energy 2017;210:1363–7.
[39] Liu H, Hu Z, Song Y, Lin J. Decentralized vehicle-to-grid control for primary frequency regulation considering charging demands. IEEE Trans Power Syst 2013;28(3):3480–9.
[40] Meng J, Mu Y, Jia H, Wu J, Yu X, Qu B. Dynamic frequency response from electric vehicles considering travelling behavior in the Great Britain power system. Appl Energy Jan. 2016;162(2):966–79.
[41] Zhu X, Xia M, Chiang HD, et al. Coordinated sectional droop charging control for EV aggregator enhancing frequency stability of microgrid with high penetration of renewable energy sources. Appl Energy Jan. 2018;210:936–43.
[42] Liu H, Hu Z, Song Y, Wang J, Xie X. Vehicle-to-grid control for supplementary frequency regulation considering charging demands. IEEE Trans Power Syst 2015;30(6):3110–9.
[43] Zhong J, He L, Li C, et al. Coordinated control for large-scale EV charging facilities and energy storage devices participating in frequency regulation. Appl Energy 2014;123(12):253–62.
[44] Peng C, Zou J, Lian L, et al. An optimal dispatching strategy for V2G aggregator participating in supplementary frequency regulation considering EV driving demand and aggregator's benefits. Appl Energy Mar. 2017;190:591–9.
[45] Falahati S, Taher SA, Shahidehpour M. Grid frequency control with electric vehicles by using of an optimized fuzzy controller. Appl Energy 2016;178:918–28.
[46] Teng F, Mu Y, Jia H, et al. Challenges on primary frequency control and potential solution from EVs in the future GB electricity system. Appl Energy 2016;194:353–62.
[47] Kaur R, Kumar ED. Transient stability analysis of IEEE 9 Bus System in power world simulator. J Eng Res Appl 2016;6(1):35–9.